

Investigating the sleep behaviours of international equestrian event riders

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RATIONALE Sleep is a fundamental behaviour for human functioning, with wide-reaching downstream consequences for health, wellbeing, and performance (Grandner, 2017). Eventing is a sport that demands optimal cognitive functioning for performance and safety. Despite the relationship between sleep and cognitive performance (Fullagar et al., 2015), there is no existing research on sleep in international equestrian athletes. Therefore, this study aimed to investigate sleep in eventers.

METHODS The study consisted of two separate components that received institutional ethical approval (ETHICS2022-151).



Figure 1. Visual summary of methods for actigraphy component

Sleep in-competition was assessed in participants ($M_{age} = 26.6$ years, $SD_{age} = 8.5$) using wrist-worn actigraphy (GENEActiv, Activinsights; Figure 1). Raw accelerometry data were analysed using the GGIR R-package to extract sleep-related variables including nocturnal total sleep time (van Hees et al., 2015). Differences between nights and gender were assessed using Friedman and Mann-Whitney U non-parametric tests, respectively.

RESULTS Participant characteristic questions highlighted unusual sleep environments, with 90% sleeping in horsebox lorries, 45% within a shared space, and 60% sharing sleeping space with a pet.

Average total sleep time during the competition was 6.1 ± 1.1 h. A reduction in sleep time on Saturday night was observed (5.3 ± 1.4 h), but there were no significant between-night differences ($p = 0.06$). A significant difference in total sleep time between male and female riders was observed ($Z = 4.17, p < .001$).

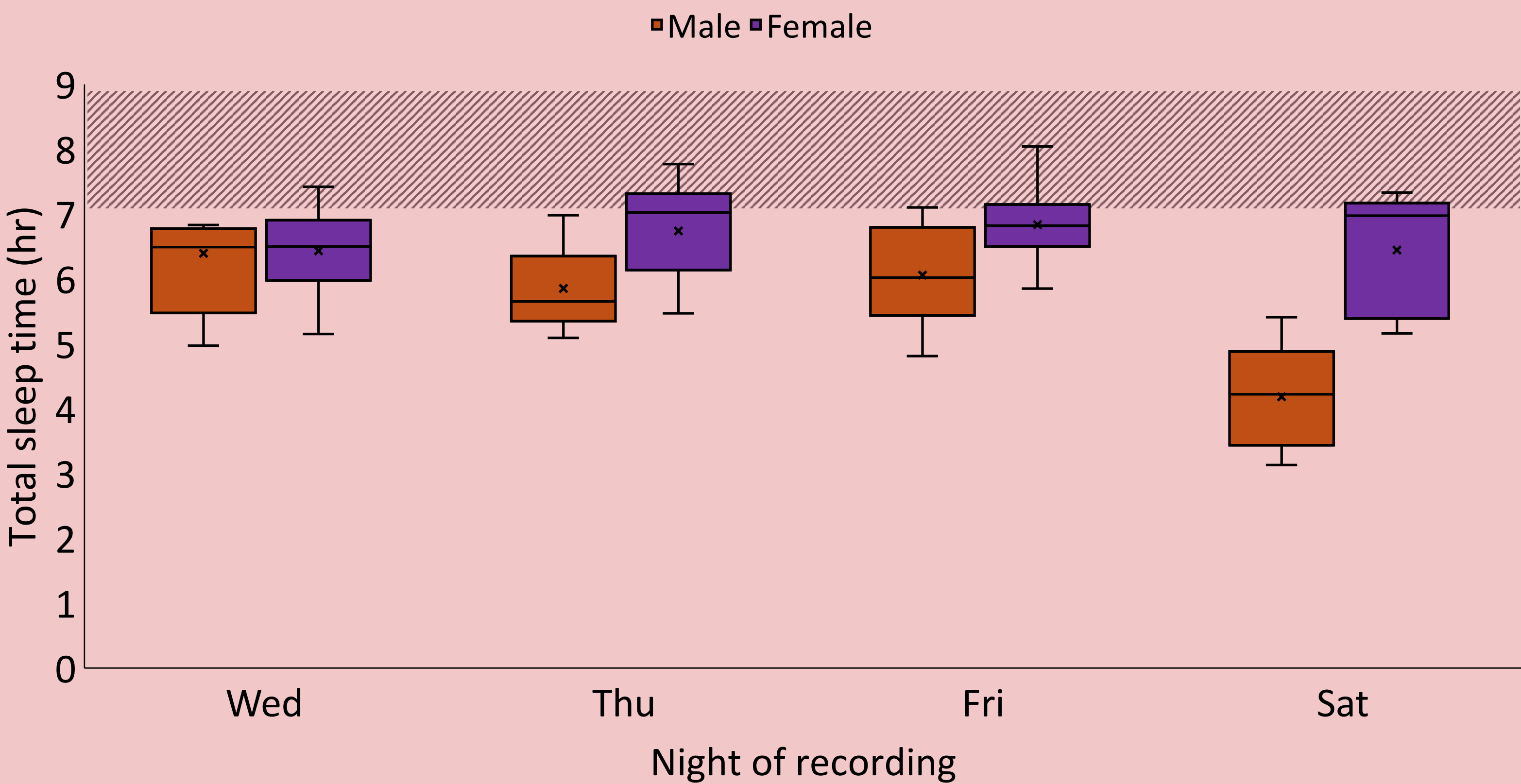


Figure 2. Boxplot displaying the distribution of total sleep time during competition split by night and gender. X denotes mean value. Dashed area correspond to health recommendations (7-9hr)

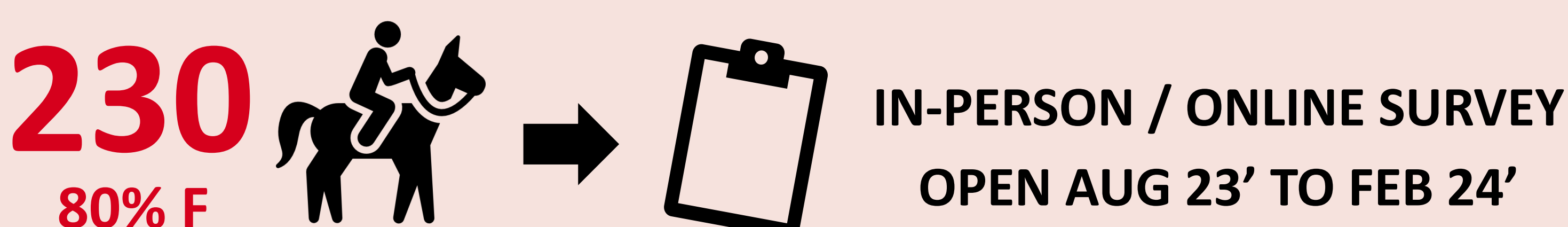


Figure 3. Visual summary of methods for survey component

The Athlete Sleep Screening Questionnaire (ASSQ) was used as a validated screening tool (Samuels et al, 2016) than can cluster sleep difficulty into four groups: none, moderate, mild, and severe. Participants (18-24 years = 43%, 25-29 years = 18%, ≥ 30 years = 39%) completed the ASSQ, demographics, and perceived facilitators and barriers to sleep in a survey using open-ended questions (Figure 3).

RESULTS ASSQ sleep difficulty groupings are presented (Figure 4). In addition, many participants reported a typical sleep duration of 5-6 hours (23%) or <5 hours (4%). Figure 5 highlights perceived facilitators and barriers to sleep at-home and in-competition.

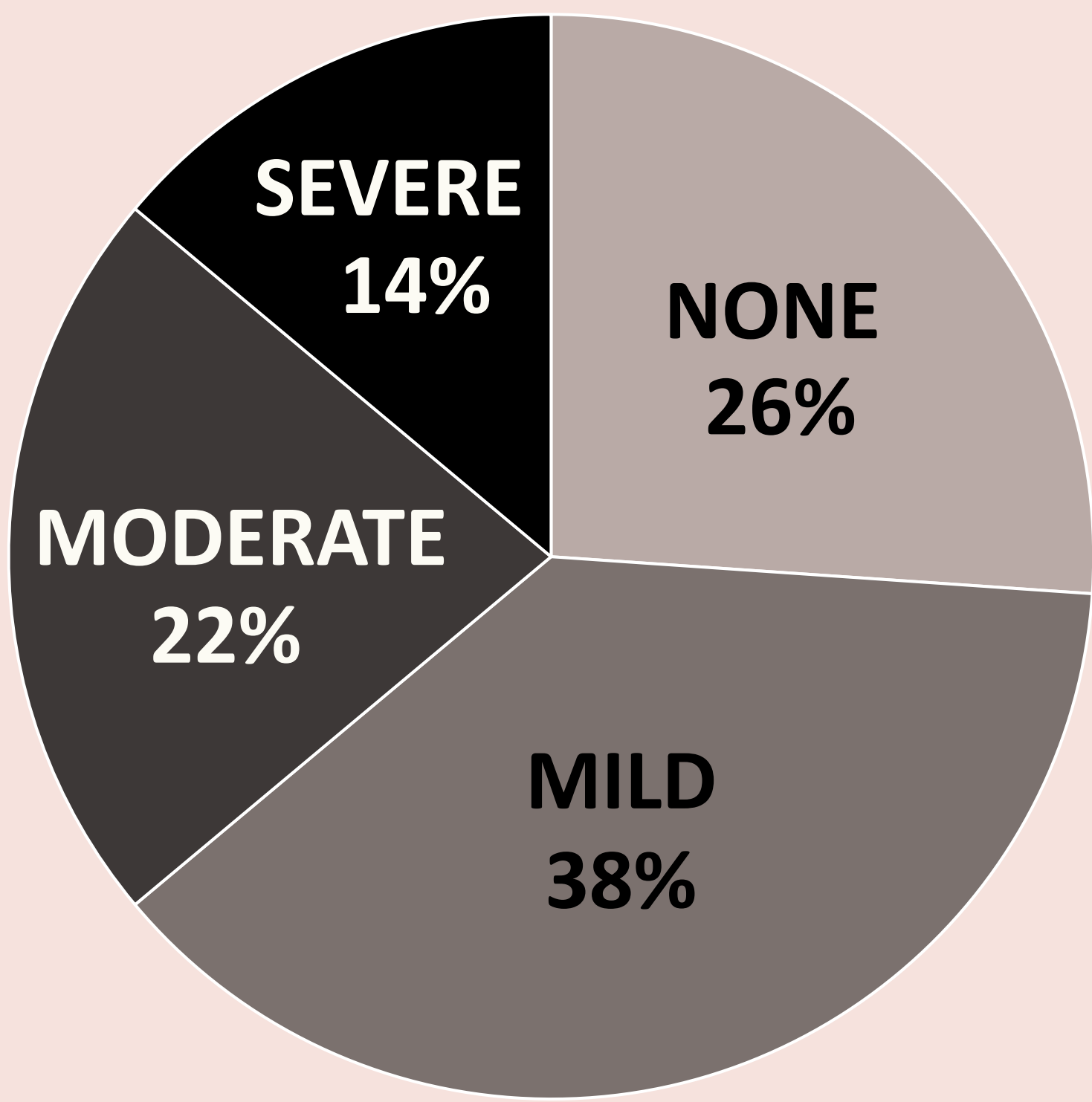


Figure 4. ASSQ sleep difficulty score frequency

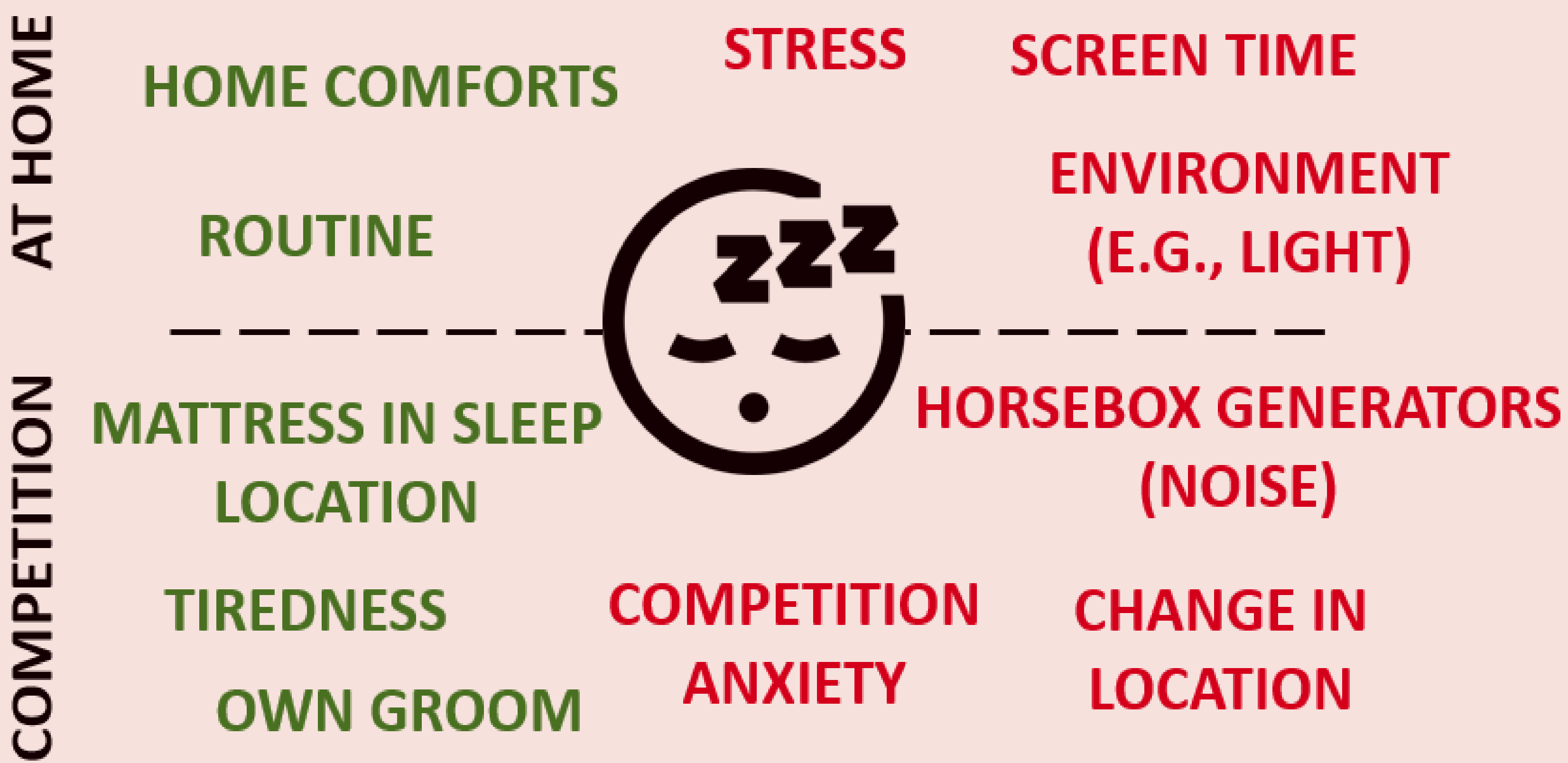


Figure 5. Reported facilitators and barriers to sleep identified through content analysis

SUMMARY Results indicate poor sleep among international event riders both at home and in-competition, such as sleep durations below health recommendations (<7 hours; Hirshkowitz et al., 2015). Notably, the ASSQ demonstrated greater sleep difficulty scores than observed in other sports (Biggins et al., 2021; Mason et al., 2023). There is a need to understand the sleep risk factors faced by eventers, and how these can be modified to support healthy sleep practices.

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